THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A habitat is a place that provides: 1

(a) the right conditions for an organism to live and grow
(b) only shelter from rain
(c) only water
(d) only food

2. Which is a biotic component? 1

(a) bacteria **(b)** temperature **(c)** sunlight **(d)** water

3. Producers are: 1

(a) bacteria that cause disease **(b)** green plants that make their own food
(c) fungi **(d)** animals

4. Decomposers include: 1

(a) grass and trees **(b)** fish and frogs
(c) lions and tigers **(d)** bacteria and fungi

5. Interconnected food chains form a: 1

(a) biome **(b)** habitat **(c)** food web **(d)** pyramid only

6. Positions in a food chain are called: 1

(a) niches only **(b)** habitats **(c)** trophic levels **(d)** biomes

7. A bee collects nectar and pollinates flowers. This is: 1

(a) commensalism **(b)** mutualism **(c)** parasitism **(d)** predation

8. Which is a human-made ecosystem? 1

(a) farm **(b)** mangrove **(c)** natural forest **(d)** coral reef

9. The Sundarbans have the world's largest: 1

(a) deserts **(b)** coral reefs **(c)** glaciers **(d)** mangrove forests

10. Invasive species harm ecosystems because they: 1

(a) fix nitrogen **(b)** purify water
(c) increase biodiversity **(d)** compete with native species

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define habitat and ecosystem. 2

12. Give two examples each of biotic and abiotic components. 2

13. Name the three kinds of organisms in an ecosystem according to their food roles. 2

14. What is a food chain? Give one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Differentiate producers, consumers and decomposers with examples. 3

16. Draw a food chain with four trophic levels and label them. 3

17. Explain why arrows in a food chain point from the eaten organism to the eater. 3

OR

Explain with an example how a change in one population affects others. 3

18. Explain how decomposers recycle nutrients. 3

OR

Compare mutualism, commensalism and parasitism in a table with one example each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain biotic and abiotic components, and describe how they interact in a pond ecosystem. 5

OR

Describe food chain and food web. Draw a food web with at least six organisms. 5

20. Explain trophic levels and the pyramid. Describe energy flow from producers to top consumers. 5

OR

Describe the role of decomposers with the activity of mushroom or compost. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND